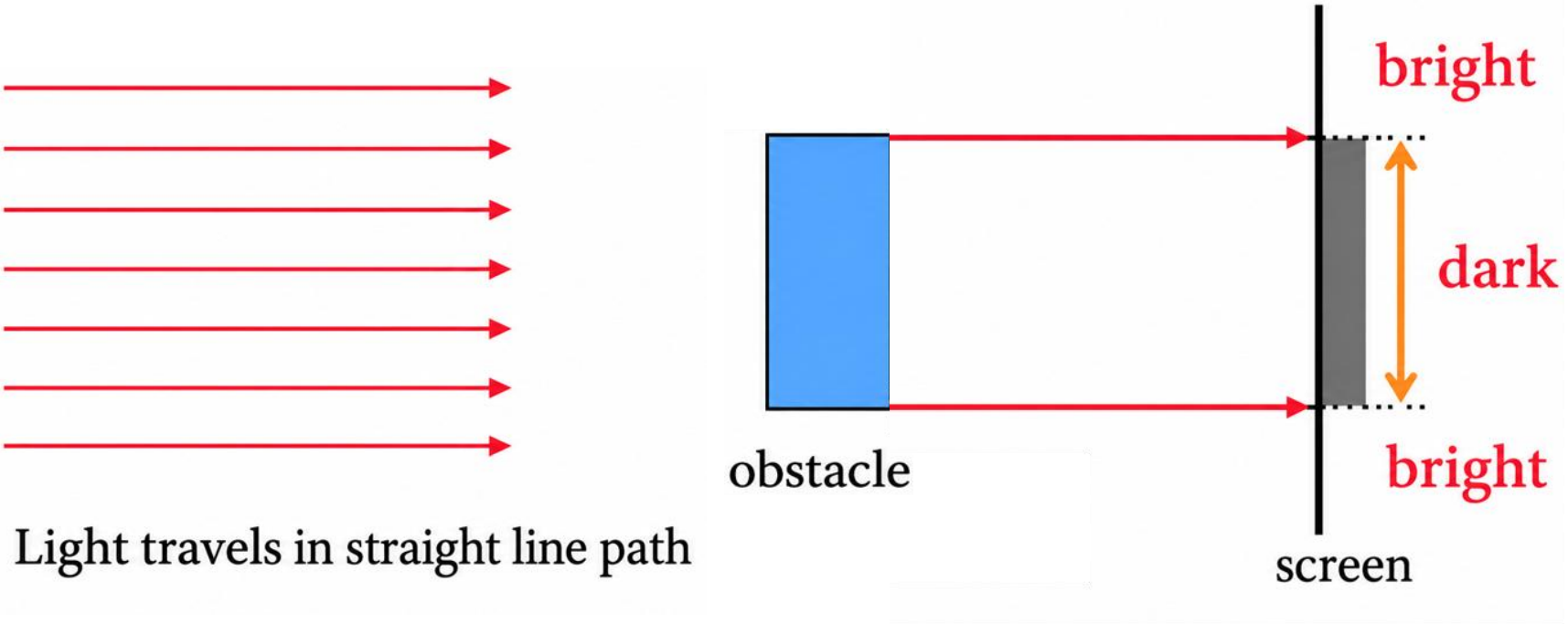
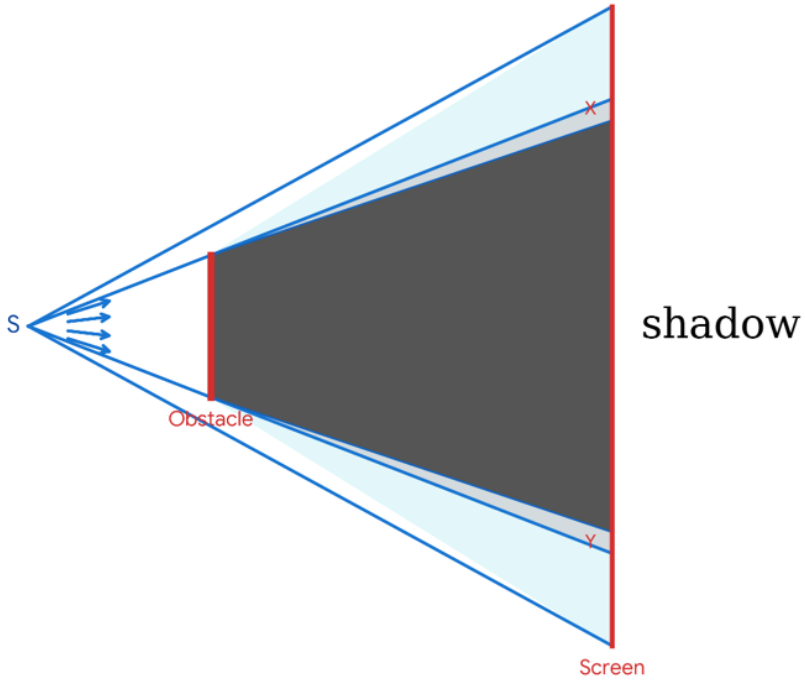
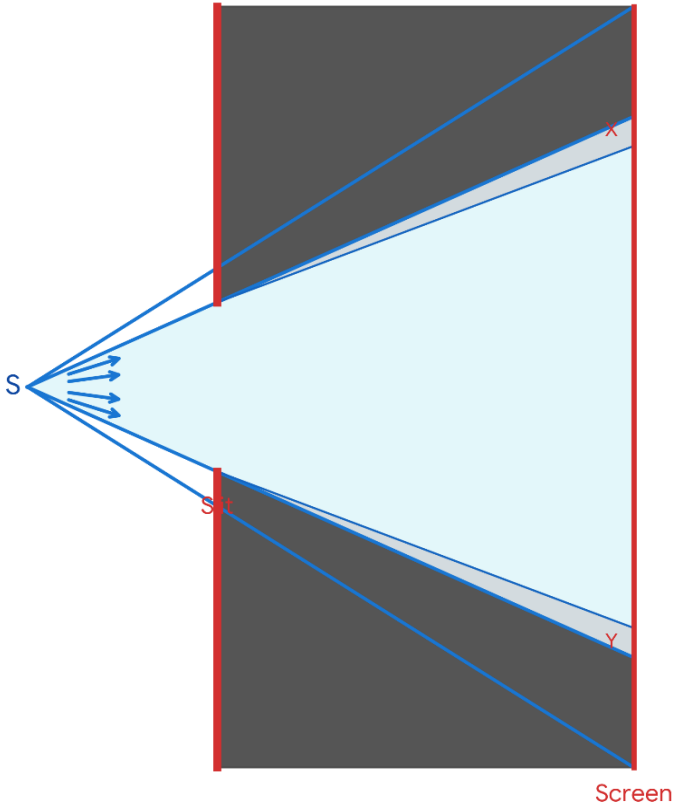
Diffraction of light



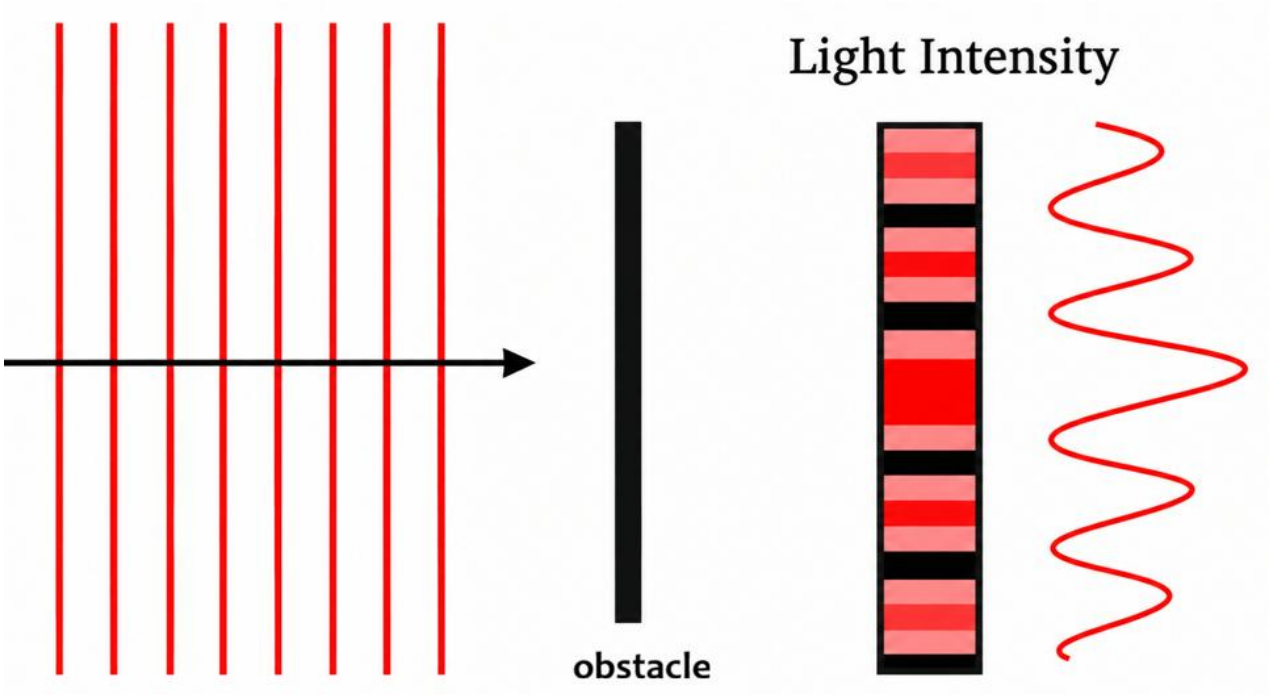


shadow



- **But in reality, the size of shadow is different than that of obstacle.**
- **This is possible due to bending of light around the corners.**

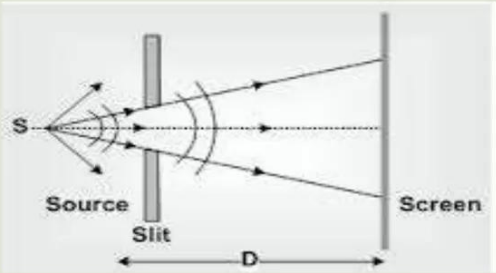
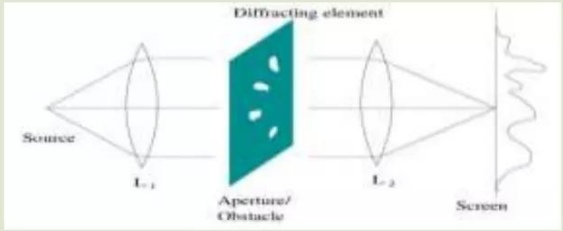
Scientist , Grimaldi found that when size of obstacle is of the order of wavelength of light, maxima and minima are obtained within the region of the geometrical shadow.



The phenomenon of bending of light around the corners of the obstacle and formation of maxima and minima within the region of the geometrical shadow is called Diffraction.

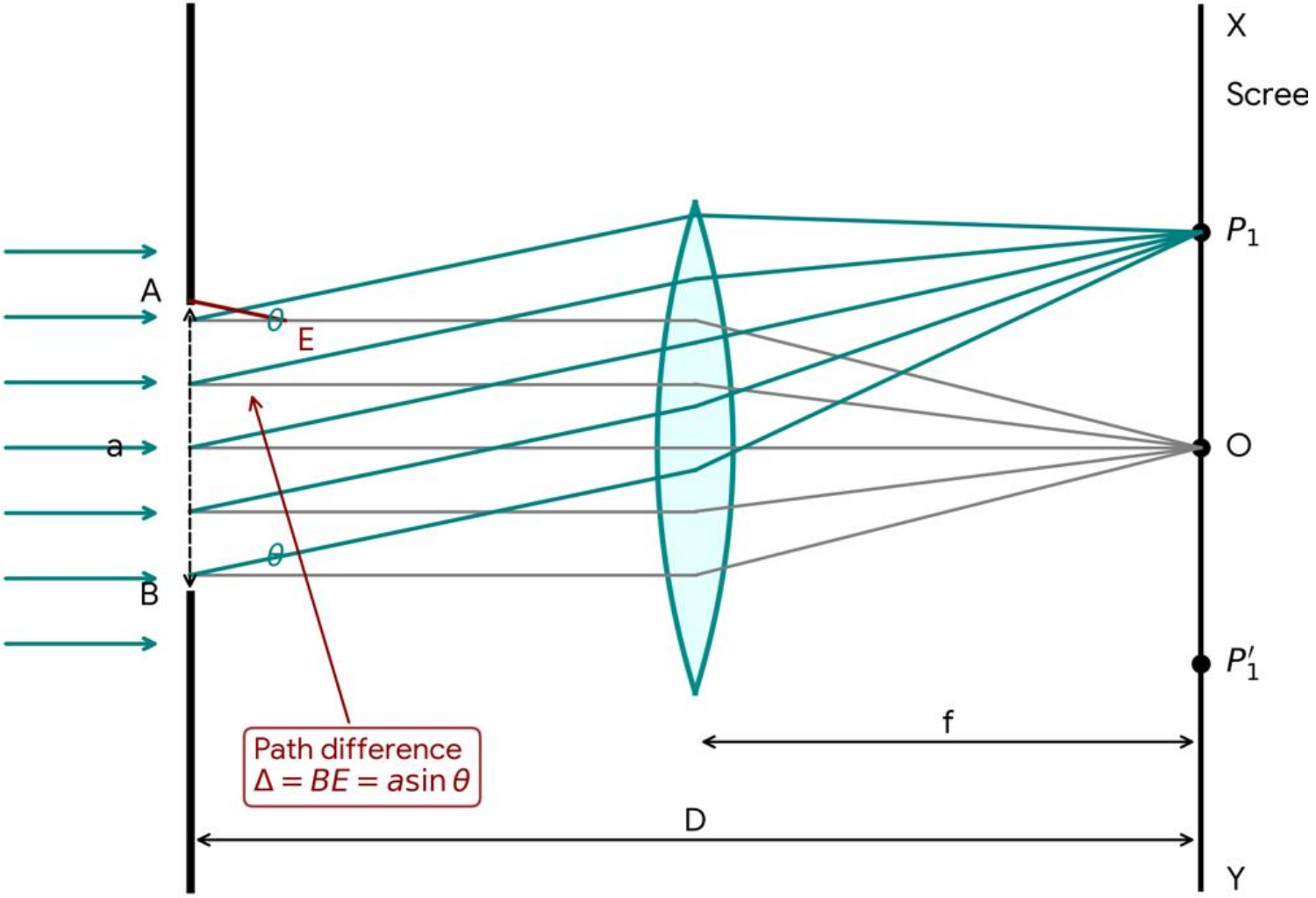
Classification of Diffraction

Diffraction phenomena of light can be divided into two different classes

Fresnel's Diffraction	Fraunhofer diffraction
	
Cylindrical wave fronts	Planar wave fronts
Source of screen at finite distance from the obstacle	Observation distance is infinite. In practice, often at focal point of lens.
Move in a way that directly corresponds with any shift in the object.	Fixed in position
Fresnel diffraction patterns on flat surfaces	Fraunhofer diffraction patterns on spherical surfaces.
Change as we propagate them further 'downstream' of the source of scattering	Shape and intensity of a Fraunhofer diffraction pattern stay constant.

https://acesse.one/ubpa1bc

Fraunhofer Diffraction at a Single Slit



For minima,

$\sin\theta_1 = \lambda/a$ (for first minima/dark)

$\sin\theta_2 = 2\lambda/a$ (for second minima/dark)

$\sin\theta_3 = 3\lambda/a$ (for third minima/dark)

and so on ……

$$a \sin\theta = n\lambda \text{ (for minima)}$$

It is natural to expect, maxima /bright half- way between two minima

$\sin\theta_1 = 3\lambda/2a$ (for first maxima/bright)

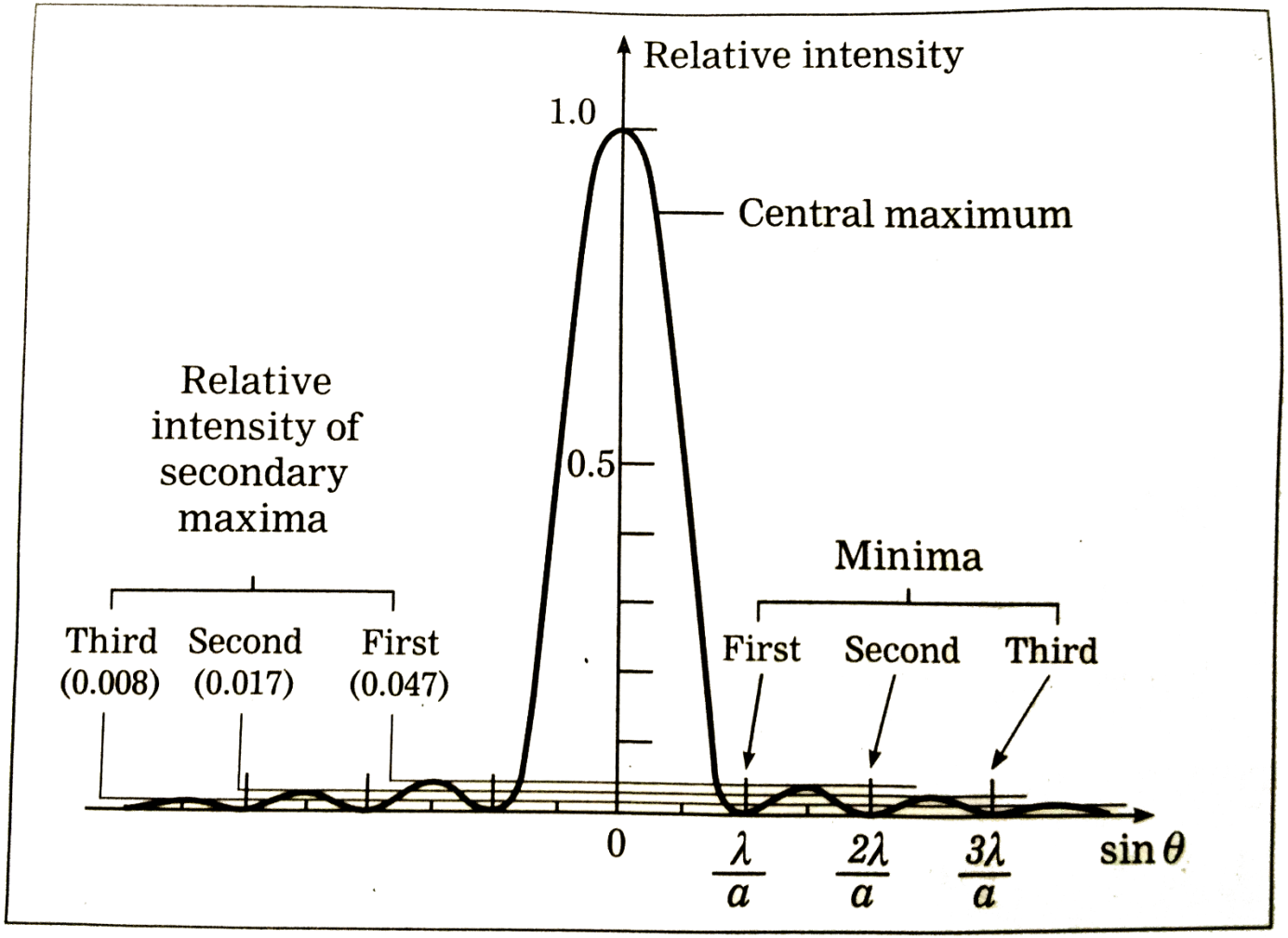
$\sin\theta_2 = 5\lambda/2a$ (for second maxima/bright)

$\sin\theta_3 = 7\lambda/2a$ (for third maxima/bright)

and so on……

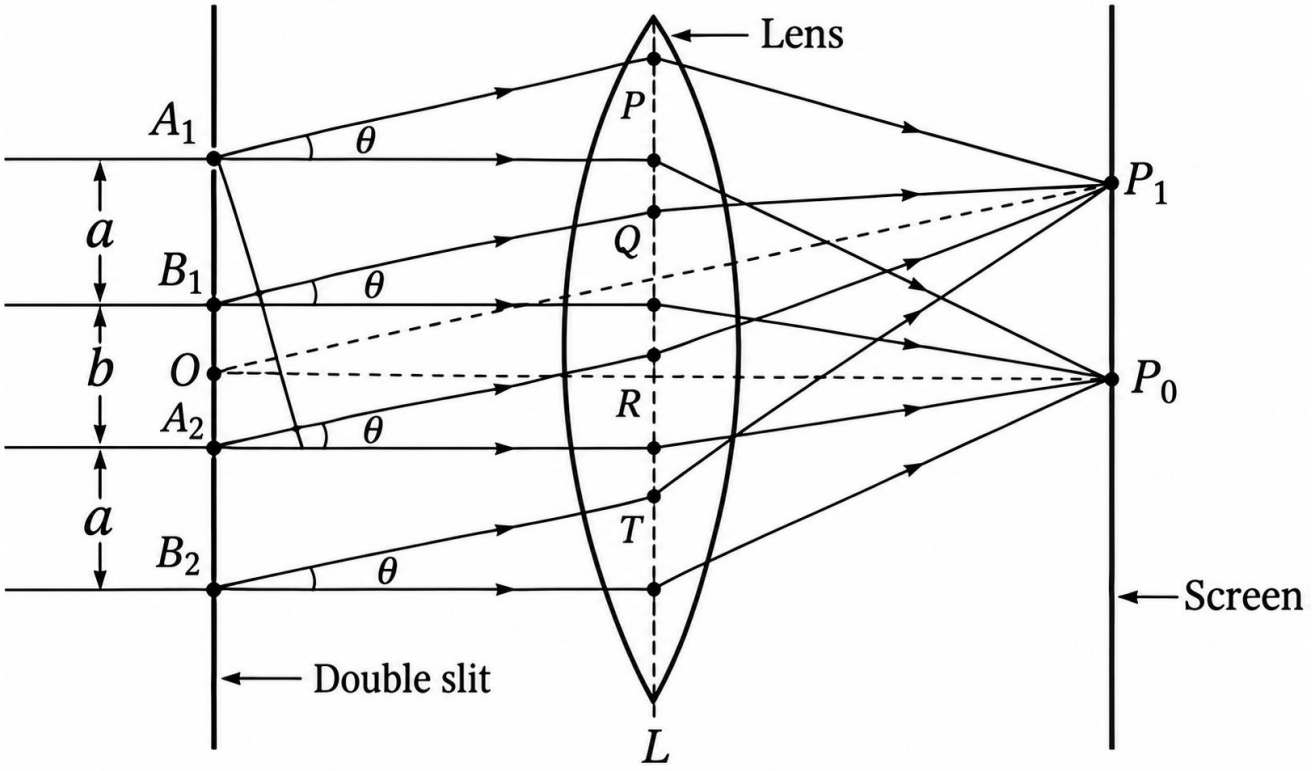
We can write, $\sin\theta_n = (2n+1)\lambda/2a$ (for nth maxima/bright)

$$a \sin\theta = (2n+1)\lambda/2 \text{ (for maxima)}$$



Intensity variation in a single slit diffraction pattern plotted against sin θ, where θ is the angular spread from the centre of the pattern

Fraunhofer Diffraction at a Double Slit



It involves both diffraction from individual slits and interference between the waves from the two slits. The resulting intensity pattern is a combination of these two phenomena.

The condition for interference maxima is given by

$$(a+b)\sin\theta = n\lambda$$

where n represents the order of the maxima

The condition for interference minima is given by

$$(a+b)\ \text{Sin}\ \theta = (2n+1)\lambda/2$$

where, n represents the order of the minima

The condition for diffraction maxima is given by

$$a\ \text{Sin}\ \theta = (2m+1)\lambda/2$$

where, m represents the order of the maxima

The condition for diffraction minima is given by

$$a\sin\theta = m\lambda$$

where, m represents the order of the minima

Diffraction Grating



Grating



Diffraction Grating

- Diffraction grating consist of large number of parallel slits of same width separated by equal opaque spacing.

- If ‘a’ is with of slit and ‘b’ is width of opaque spacing, then (a+b) is called grating element or grating constant.

- N is number of lines per cm on the grating, then
  (a+b)N = 1 cm

The grating constant (a+b) and the number of lines per centimeter (N) on a diffraction grating are inversely related by the formula

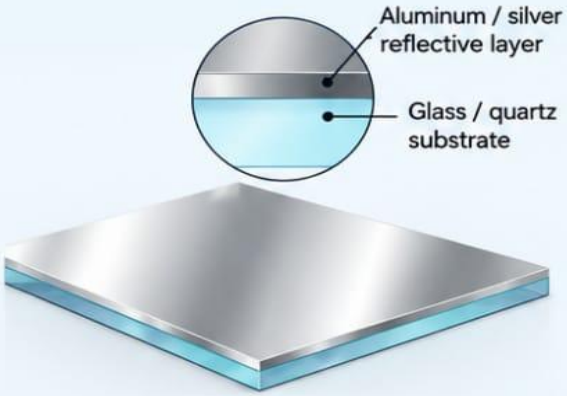
$$(a+b) = 1/N$$

For example, a grating with 5000 lines per centimeter would have a grating constant of 1/5000 cm.

- A plane transmission grating is constructed by ruling a large number of parallel, equidistant, and extremely fine lines on optically flat glass plate using a precise, diamond-pointed ruling engine.
- The lines drawn by the diamond scratch the glass surface, making it opaque. These opaque scratches act as obstacles. The undisturbed, clear glass spaces between these scratches remain transparent and act as fine slits.
- The commercial gratings are usually replicas. They are made by pouring a thin layer of liquid cellulose acetate or resin onto the master grating. Once hardened, this film is stripped off, mounted between two flat glass plates, and used as a transmission grating.

How a Diffraction Grating is Made

1 An optically flat glass or quartz plate is polished to extreme precision and vapor-coated with a highly uniform layer of reflective aluminum or silver.



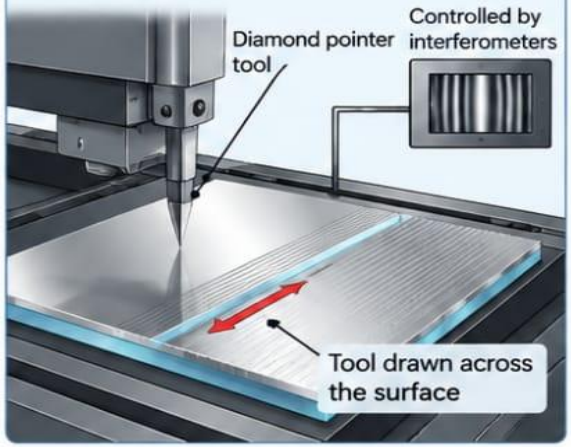
Glass/quartz plate with reflective aluminum/silver coating

2 This plate is placed into an automated machine called a ruling engine.

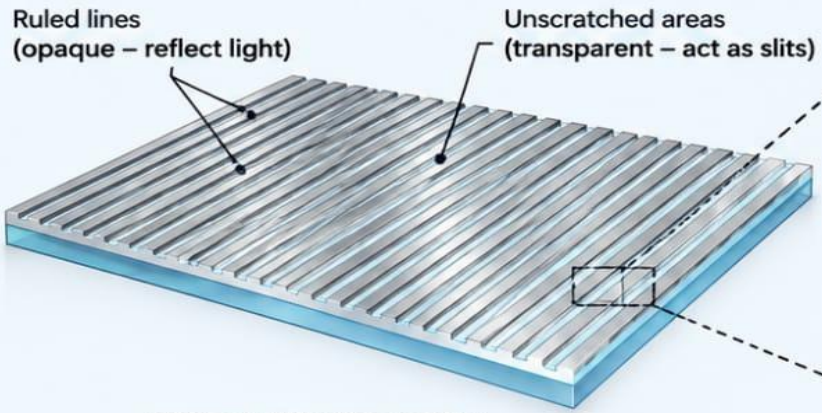


Ruling engine

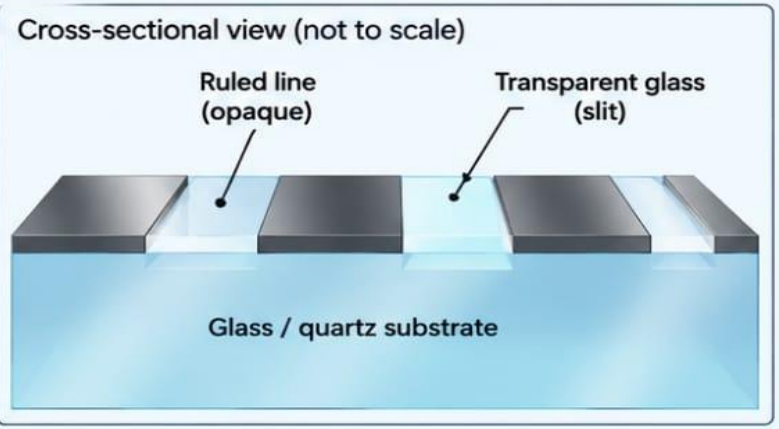
3 Controlled by interferometers to avoid micro-scale errors, a precise, sharp diamond pointer tool is physically drawn across the metal film equal to number of lines required times for every inch.



4 The diamond pointer carves out precise grooves; the unscratched transparent glass acts as the slits, while the ruled lines act as opaque barriers to light.



Finished diffraction grating



- The working of a diffraction grating is based on the combined phenomena of diffraction (the bending of light around obstacles) and interference (the superposition of waves from different slits)

- A parallel beam of monochromatic light of wavelength λ is made to fall normally on the surface of the grating

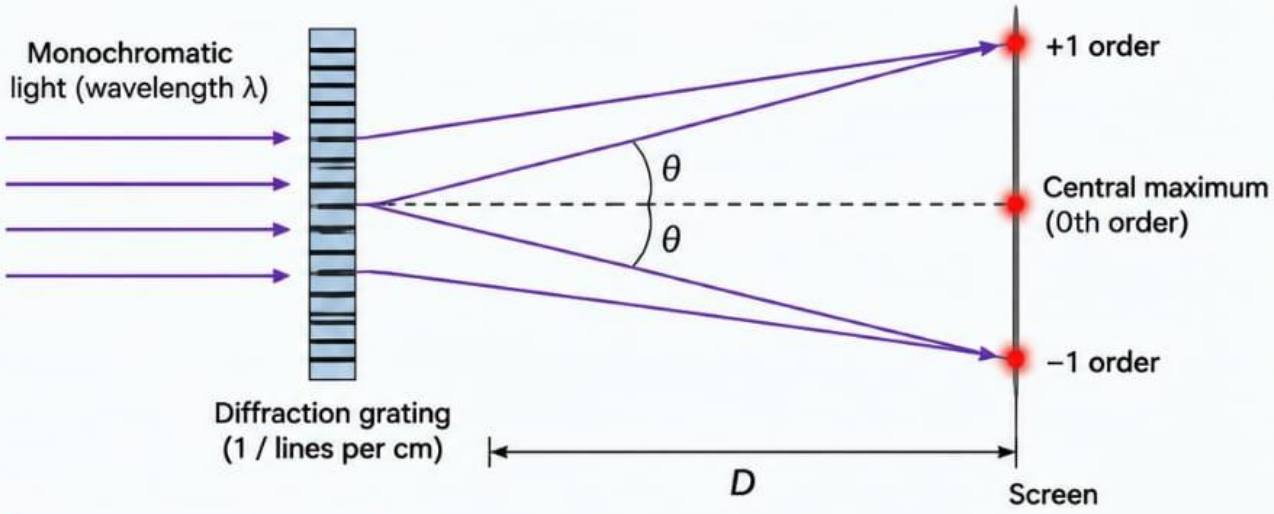
- When monochromatic parallel light falls normally on the grating, the light passes through all the transparent slits.

- Each slit produces a diffracted wave, and these waves overlap and interfere with one another.

- At certain angles, the waves from all the slits reach the screen in phase $\rightarrow$ bright principal maximum.

- At other angles, they reach with different phases and largely cancel $\rightarrow$ dark regions

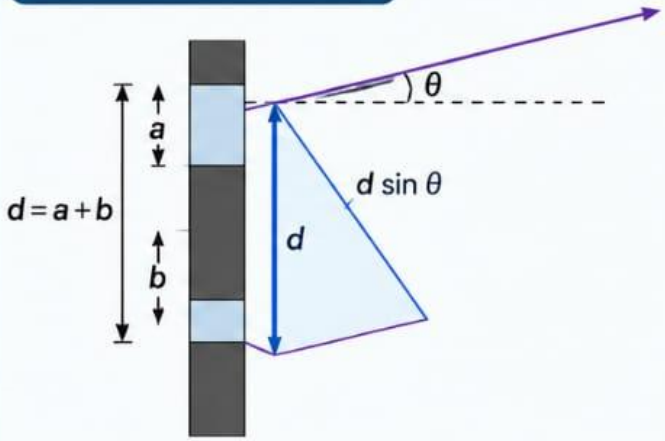
Diffraction Grating – Derivation of Grating Formula (a + b)



a	=	width of transparent slit
b	=	width of opaque part (ruled line)
d	=	grating element $(a + b)$
λ	=	wavelength of light
θ	=	angle of diffraction
D	=	distance from grating to screen
n	=	order of maximum (n = 1, 2, 3, ...)

Number of lines per cm, $N = \frac{1}{d}$
(if d is in cm)

Derivation of a + b



For constructive interference (principal maxima):

$$d \sin\theta = n\lambda$$

Since $d = a + b$,

$$(a + b) \sin\theta = n\lambda$$

where,

n = 1, 2, 3, ... (order of maximum)

d = grating element $(a + b)$

Grating formula:

$$a + b = \frac{n\lambda}{\sin\theta}$$

(or)

$$N = \frac{1}{d} = \frac{\sin\theta}{n\lambda}$$

where N = number of lines/cm

The diffraction grating formula is $(a + b) \sin \theta = n \lambda$

where (a + b) represents the grating element (the combined width of a transparent slit and an opaque space)

a: Width of the transparent slit.

b: Width of the opaque region (width of the obstacle between slits).

(a + b) or (d): Grating element (the distance between corresponding points of adjacent slits).

θ: Angle of diffraction (the angle between the direction of the diffracted beam and the normal to the grating).

n: Order of the maximum diffraction (an integer: 0, 1, 2, 3,…..)

λ: Wavelength of the incident light

Resolving Power

1. When two objects/wavelengths are very close to each other or far away from us we **NOT BE ABLE** to distinguish.

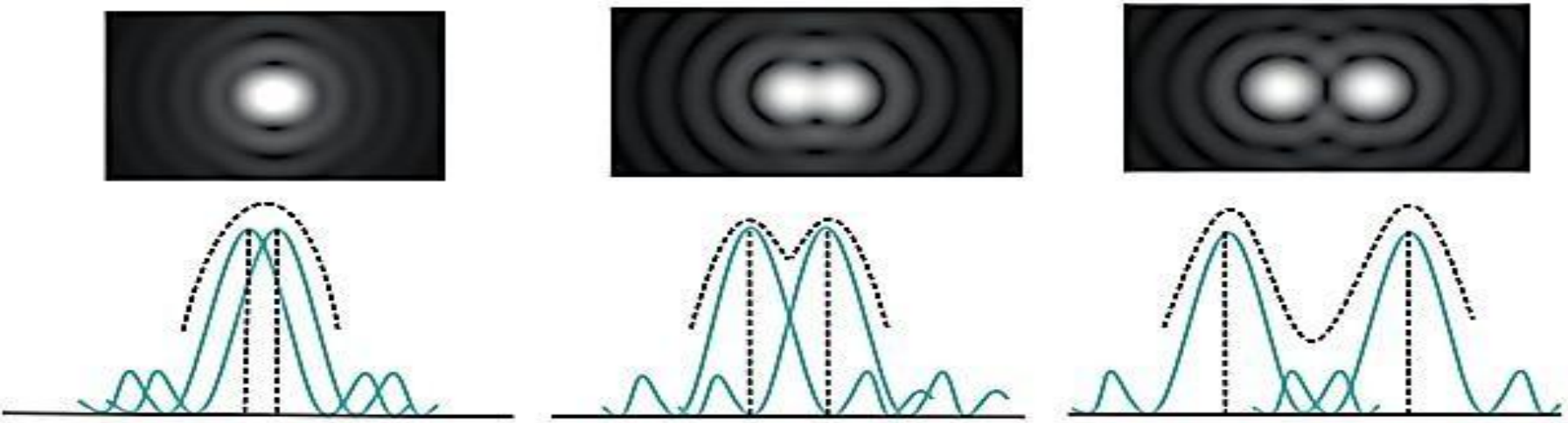
2. To distinguish them we need optical instruments like microscope, telescope, grating etc.

3. The resolving power of an objective lens is measured by its ability to differentiate two object/ wavelengths. The greater the resolving power, the smaller the minimum distance between two lines or points that can still be distinguished.

Rayleigh's Criterion for Limit of Resolution

1. According to Rayleigh, the ability of an optical instrument to distinguish between two closely spaced objects depends upon the diffraction patterns of the two objects.

2. According to this criterion, two objects are just resolved when the first minimum of the diffraction pattern of one source coincides with the central maximum of the diffraction pattern of the other source, and vice versa.



(a) Unresolved (b) Just resolved (c) well resolved

Resolving Power of a grating

The spectral resolving power of a diffraction grating is its ability to distinguish between two closely spaced wavelengths and is defined as the ratio of the average wavelength (λ) to the smallest measurable difference in wavelength (dλ),

i.e., **R.P = λ/dλ**

Sodium vapor lamps emit a famous bright yellow light which actually consists of two very closely spaced wavelengths: $\lambda_1 = 589.0\ \text{nm}$ and $\lambda_2 = 589.6\ \text{nm}$.

- **Step 1: Find the average wavelength (λ) and the difference ($\Delta\lambda$).**

$$\lambda = \frac{589.0 + 589.6}{2} = 589.3\ \text{nm}$$

$$\Delta\lambda = 589.6 - 589.0 = 0.6\ \text{nm}$$

- **Step 2: Calculate the required resolving power (R).**

$$R = \frac{\lambda}{\Delta\lambda} = \frac{589.3\ \text{nm}}{0.6\ \text{nm}} \approx 982$$

Astronomers use massive gratings to detect the "Zeeman effect" (the splitting of spectral lines due to a star's magnetic field). Suppose an astronomer needs to split an iron absorption line at $\boldsymbol{\lambda = 630.2\ \mathbf{nm}}$ where the split components are separated by a mere $\boldsymbol{\Delta\lambda = 0.015\ \mathbf{nm}}$.

Step 1: Calculate the required resolving power (R).

$$R = \frac{\lambda}{\Delta\lambda} = \frac{630.2\ \text{nm}}{0.015\ \text{nm}} \approx 42,013$$

The resolving power is directly proportional to the order of the spectrum (n) and the total number of lines (N') on the grating

The resolvance or chromatic resolving power of the grating, expressed as

$$\mathbf{R.P = nN'}$$

If we use a standard grating with 15000 lines per cm, then only a tiny width of about 6.667×10^{-5} cm **will be seen in the first order**

- Two wavelengths are well resolved

$$\lambda/d\lambda < nN'$$

- Two wavelengths are just resolved

$$\lambda/d\lambda = nN'$$

- Two wavelengths are not resolved

$$\lambda/d\lambda > nN'$$

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- **Step 2: Calculate the required resolving power (R).**

$$R = \frac{\lambda}{\Delta\lambda} = \frac{589.3 \text{ nm}}{0.6 \text{ nm}} \approx 982$$

To separate these two yellow lines in the **first-order spectrum (n = 1)**, the grating must have at least (982) lines (N = R/n).

Astronomers use massive gratings to detect the "Zeeman effect" (the splitting of spectral lines due to a star's magnetic field). Suppose an astronomer needs to split an iron absorption line at $\lambda = 630.2\ \text{nm}$ where the split components are separated by a mere $\Delta\lambda = 0.015\ \text{nm}$.

Step 1: Calculate the required resolving power (R).

$$R = \frac{\lambda}{\Delta\lambda} = \frac{630.2\ \text{nm}}{0.015\ \text{nm}} \approx 42,013$$

If the astronomer observes this in the **third-order ($n = 3$)**, the number of illuminated lines needed is:

$$N = \frac{R}{n} = \frac{42,013}{3} \approx 14,004\ \text{lines}$$

- Two wavelengths are well resolved

$$\lambda/d\lambda < nN'$$

- Two wavelengths are just resolved

$$\lambda/d\lambda = nN'$$

- Two wavelengths are not resolved

$$\lambda/d\lambda > nN'$$